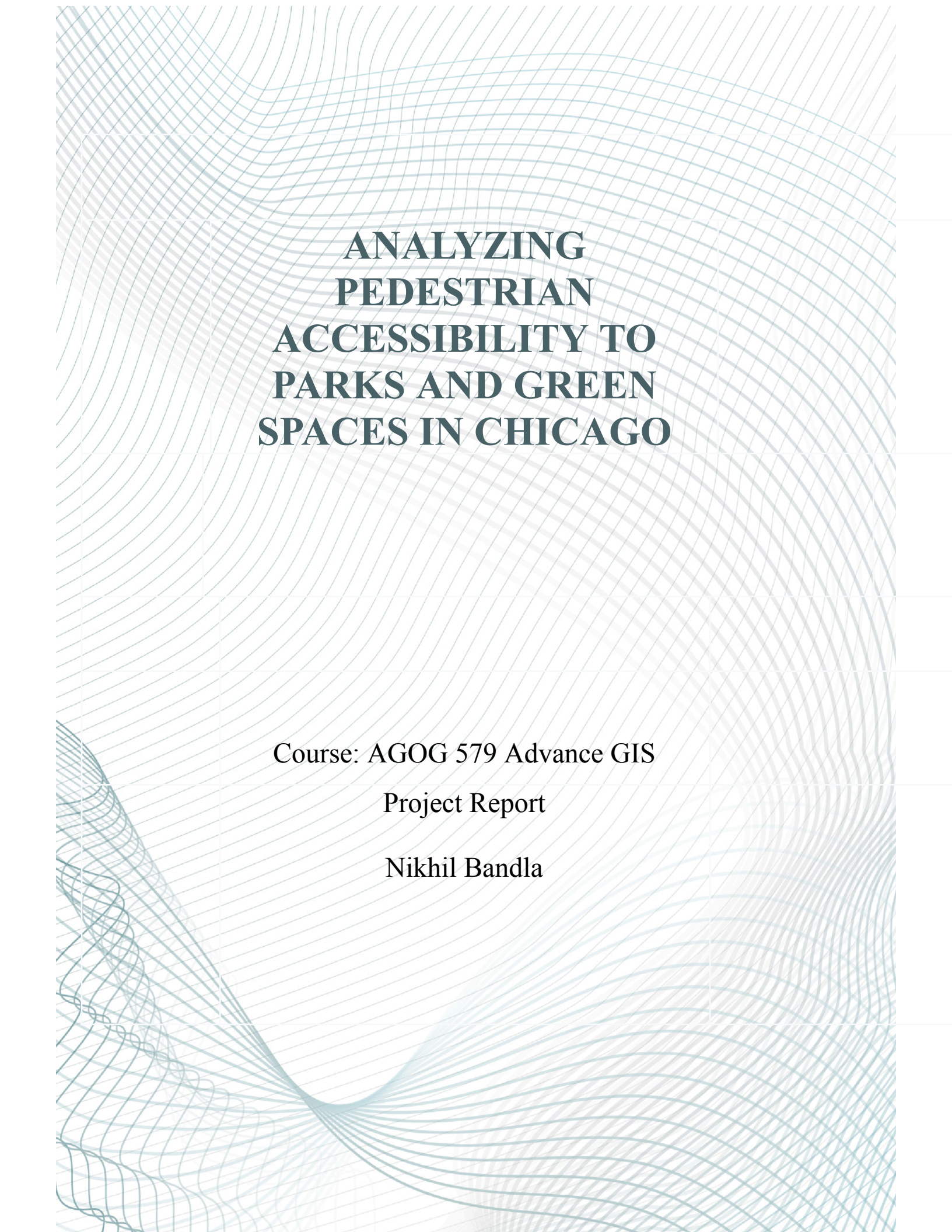




ANALYZING PEDESTRIAN ACCESSIBILITY TO PARKS AND GREEN SPACES IN CHICAGO

Course: AGOG 579 Advance GIS

Project Report

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Analysing Pedestrian Accessibility to Parks and Green Spaces in Chicago

Abstract

Urban parks and green spaces are vital components of healthy, livable cities, yet access to these resources remains inequitably distributed across urban populations. This study evaluates pedestrian accessibility to parks and green spaces in Chicago, Illinois, using Geographic Information Systems (GIS) to compare two analytical methods: buffer analysis and network analysis. Buffer zones of 400 meters and 800 meters were created around all park features to measure straight-line proximity, while network analysis calculated realistic walking distances along the street network. Results demonstrate significant disparities between the two approaches. The 800-meter buffer analysis indicated that 9.22% of Chicago's land area (594.4 km²) falls within walking distance of parks, potentially serving approximately 251,000 residents. However, network analysis revealed that only 5.78% of the city (372.5 km²) has actual pedestrian access within 800 meters of walking distance via streets, serving an estimated 157,000 residents. This represents a 37% reduction in accessibility when accounting for real-world pedestrian infrastructure. The network-based 800-meter walking distance produced nearly identical results to the 400-meter straight-line buffer, suggesting that actual pedestrian routes in Chicago are approximately double the straight-line distance. These findings emphasize the critical importance of considering street network configuration and pedestrian infrastructure when evaluating urban green space accessibility. The research provides evidence-based insights for urban planning initiatives aimed at promoting equitable park access and improving public health outcomes through enhanced walkability to green spaces.

Keywords: urban parks, pedestrian accessibility, GIS, network analysis, buffer analysis, Chicago, environmental justice.

Introduction

General Overview of Urban Green Spaces

Urban parks and green spaces constitute essential elements of sustainable and liveable cities, serving multiple critical functions that extend far beyond aesthetic enhancement. These spaces encompass diverse landscape types, including municipal parks, community gardens, urban forests, greenways, and recreational areas that provide natural environments within built urban contexts. As urbanization intensifies globally, with over 55% of the world's population now residing in cities, the role of accessible green spaces in supporting human health, environmental quality, and social equity has become increasingly prominent in urban planning discourse.

Green spaces deliver substantial environmental benefits to urban ecosystems. They mitigate urban heat island effects by providing shade and facilitating evapotranspiration, which can reduce ambient temperatures by several degrees Celsius during summer months. Urban vegetation acts as a natural air filter, absorbing pollutants such as nitrogen dioxide, sulphur dioxide, and particulate matter while producing oxygen through photosynthesis. Parks and green corridors support urban biodiversity by providing habitat for birds, insects, and other wildlife species, maintaining ecological connectivity within fragmented urban landscapes. Additionally, permeable green surfaces facilitate stormwater infiltration, reducing runoff volume and flood risks while recharging groundwater supplies.

The public health benefits of urban green spaces are well-documented across multiple domains. Regular access to parks encourages physical activity, including walking, jogging, cycling, and sports participation, helping combat sedentary lifestyles and associated health conditions such as obesity, cardiovascular disease, and diabetes. Exposure to natural environments has been linked to reduced stress levels, improved mental health outcomes, and enhanced cognitive function. Studies demonstrate that proximity to green spaces correlates with lower rates of depression and anxiety, improved attention restoration, and better overall psychological well-being. For children, access to parks supports healthy physical development, imaginative play, and environmental education opportunities. Social cohesion and community well-being are significantly influenced

by the availability of quality public green spaces. Parks serve as common grounds where diverse populations can interact, fostering social connections across demographic boundaries. They provide venues for community events, cultural celebrations, and informal gatherings that strengthen neighborhood identity and social capital. For marginalized populations, including low-income residents and minority communities, parks offer rare opportunities for recreation and respite that might otherwise require costly private alternatives or travel to distant locations.

Background and Context

Despite these recognized benefits, access to urban parks remains inequitably distributed across many cities. Historical patterns of urban development, discriminatory land-use policies, and socioeconomic disparities have resulted in systematic differences in green space availability among neighborhoods. Lower-income communities and minority neighborhoods frequently experience deficits in park acreage, quality, and proximity compared to affluent areas. This environmental injustice has tangible consequences for public health outcomes, with residents of park-poor neighborhoods experiencing higher rates of chronic diseases and reduced life expectancy.

Chicago presents a particularly relevant context for examining urban park accessibility. As the third-largest city in the United States, with approximately 2.7 million residents spread across 234 square miles, Chicago exhibits considerable diversity in neighborhood characteristics, socioeconomic conditions, and urban form. The city maintains over 600 parks managed by the Chicago Park District, representing a substantial public investment in green infrastructure. However, questions persist regarding whether these resources are equitably distributed and accessible to all residents, particularly those dependent on pedestrian transportation.

The spatial arrangement of Chicago's parks reflects over a century of urban development influenced by various planning philosophies, political decisions, and economic forces. While some neighborhoods benefit from historic park systems designed during the City Beautiful movement, others developed during periods of rapid industrial expansion with minimal attention to recreational space allocation. Contemporary urban planning efforts increasingly recognize the

importance of park accessibility, yet systematic evaluation of current conditions remains essential for identifying gaps and informing targeted interventions.

Problem Statement

Although Chicago possesses an extensive park system, the degree to which residents can actually access these green spaces through pedestrian travel remains unclear. Traditional approaches to measuring park accessibility often rely on simple proximity measures such as straight-line distance or circular buffer zones around park features. However, these methods fail to account for the reality of urban mobility, where physical barriers, street network configuration, and pedestrian infrastructure constraints significantly affect whether residents can realistically walk to nearby parks.

The discrepancy between theoretical proximity and functional accessibility has important implications for understanding environmental justice, public health planning, and infrastructure investment priorities. A park may appear geographically close to a residential area on a map yet remain functionally inaccessible if highways, industrial zones, or inadequate pedestrian facilities impede walking routes. Without accounting for actual pedestrian travel conditions, planners and policymakers may overestimate park accessibility and fail to recognize populations truly underserved by the existing green space network.

Project Objectives

This research aims to provide a comprehensive, evidence-based assessment of pedestrian accessibility to parks and green spaces throughout Chicago using Geographic Information Systems (GIS) analysis. The study pursues several specific objectives:

1. Quantify the spatial extent of park accessibility across Chicago using both buffer analysis (straight-line proximity) and network analysis (actual pedestrian walking distance via streets) to measure coverage at 400-meter and 800-meter distance thresholds corresponding to 5-minute and 10-minute walking times.

2. Compare the results of buffer-based and network-based accessibility measures to quantify the degree to which traditional proximity methods overestimate actual pedestrian access and to identify systematic patterns in how urban form affects walkability to parks.
3. Estimate the number of Chicago residents living within accessible walking distance of parks using population density analysis applied to the identified accessible areas.
4. Generate detailed cartographic visualizations illustrating spatial patterns of park accessibility across the city to support the interpretation and communication of findings to planning practitioners and policymakers.

This research provides insights that can inform urban planning decisions aimed at promoting environmental justice and improving public health outcomes through enhanced access to green spaces. The findings will highlight areas of the city where park access deficits are most pronounced and where investments in either new park development or improved pedestrian infrastructure could yield the greatest benefits for underserved populations.

Research Questions

This study addresses the following research questions:

1. What percentage of Chicago's total area falls within accessible pedestrian walking distance (400m and 800m) of parks and green spaces?
2. How does network-based accessibility analysis, accounting for actual street networks, compare to simple Euclidean buffer analysis?
3. What are the spatial patterns of park accessibility across Chicago, and which areas experience limited access to green spaces?

Related Works

The measurement and evaluation of urban park accessibility have been a longstanding focus in urban planning and public health research. Traditional approaches to assessing park access have evolved from simple proximity measures to sophisticated spatial analysis techniques that account for the complexities of urban mobility and transportation infrastructure.

Buffer analysis, utilizing Euclidean distance measurements, represents one of the most employed methods for evaluating park accessibility. Sister et al. (2010) applied buffer-based approaches to assess park access across major U.S. cities, establishing benchmark distances of 400 meters and 800 meters as reasonable walking thresholds for urban residents. These distance standards have been widely adopted in subsequent research and policy documents, including guidelines from the Trust for Public Land and various municipal planning departments.

However, simple buffer analysis has limitations in representing actual pedestrian accessibility. Network analysis approaches, which calculate accessibility based on walkable street networks rather than straight-line distances, provide more realistic assessments of park access. Talen and Anselin (1998) pioneered network-based accessibility measures in urban park research, demonstrating significant differences between Euclidean and network distance calculations. Their work highlighted how urban form, including street connectivity and barriers such as highways or rivers, substantially influences actual accessibility patterns.

Recent advances in GIS technology have enabled more sophisticated network analyses. Higgs et al. (2015) employed service area analysis using detailed street networks to evaluate park accessibility in urban environments, accounting for variations in street connectivity and pedestrian infrastructure. This approach provides more accurate representations of accessibility by considering the actual paths residents would traverse to reach parks.

Research specifically focusing on Chicago has revealed disparities in park access across different neighborhoods. While previous studies have documented these inequities, this research extends

existing work by employing both buffer and network analysis methods to provide a comprehensive assessment of current accessibility patterns throughout the city.

Methods

Data Acquisition

All spatial data utilized in this analysis were obtained from the Chicago Data Portal, the city's official open data repository. Three primary datasets formed the foundation of this study. The Chicago Park District Park Boundaries dataset provided polygon geometries representing the spatial extent of all parks managed by the Chicago Park District, including detailed attribute information about park characteristics and locations. The Boundaries-City dataset supplied the official municipal boundary of Chicago, enabling calculation of citywide accessibility metrics. The Street Center Lines dataset contained comprehensive line geometries representing Chicago's street network, essential for network-based accessibility analysis.

All datasets were projected to the North American Datum 1983 (NAD 1983) Universal Transverse Mercator (UTM) Zone 16N coordinate system to ensure accurate distance measurements in meters and maintain consistency across all spatial analyses. The Park Boundaries and Street Centerlines datasets were clipped to the Chicago city boundary to focus the analysis exclusively on the study area and eliminate extraneous features outside municipal limits.

Spatial Analysis Techniques

This research employed two distinct GIS methodologies to assess park accessibility, each offering different perspectives on how residents can reach green spaces. The first approach, buffer analysis, calculated zones of proximity around park features to represent potential accessibility based on straight-line distance. Using the Buffer geoprocessing tool in ArcGIS Pro, circular zones were created at two distance thresholds: 400 meters and 800 meters. These

distances correspond to approximately 5-minute and 10-minute walking times respectively, based on average pedestrian speeds of 80 meters per minute commonly used in urban planning research. The dissolve type parameter was set to "ALL" to merge overlapping buffer zones, ensuring that areas served by multiple parks were counted only once in accessibility calculations.

Following buffer creation, the Calculate Geometry tool was used to determine the total area (in square meters) covered by each buffer distance. These area values were then compared to Chicago's total land area of 6,450,276,623 square meters to calculate the percentage of the city falling within each accessibility threshold.

The second analytical approach employed network analysis to measure actual pedestrian walking accessibility via the street network. This method provides a more realistic assessment of park access by accounting for physical barriers, street layouts, and the absence of direct pedestrian connections that buffer analysis cannot capture. A network dataset was constructed from the Street Centerlines layer using ArcGIS Pro's Network Analyst extension. The network dataset enabled routing analysis that respects real-world travel paths along streets rather than assuming unobstructed straight-line movement.

Using the Service Area tool within Network Analyst, 800-meter walking distance zones were generated from park entrance points. Park polygon features were first converted to point features representing access locations. The service area analysis calculated all areas reachable within 800 meters of walking distance along the street network from these park points. The resulting service area polygons were dissolved into a single feature to eliminate overlaps and calculate total accessible area.

Population Analysis

To estimate the number of Chicago residents with access to parks, a population density approach was implemented. Chicago's total population of 2,721,326 (based on recent census estimates) was divided by the city's total land area to calculate average population density of 0.422 persons per square meter. This density value was then multiplied by the accessible areas identified

through both buffer and network analyses to estimate the number of residents living within walking distance of parks.

While this citywide density approach provides a reasonable approximation, it represents a limitation of the analysis as it does not account for variations in population distribution across different neighborhoods. More granular census block-level demographic data would provide more precise population estimates but was not incorporated in this study.

Results

Buffer Analysis Findings

The 400-meter buffer analysis revealed that 371,926,365 square meters (approximately 371.9 square kilometres) of Chicago's land area falls within this proximity threshold. This represents 5.77% of the city's total area. Applying the population density calculation, approximately 157,000 residents are estimated to live within a 5-minute straight-line walk of parks and green spaces.

The 800-meter buffer analysis demonstrated significantly greater coverage, with 594,407,811 square meters (approximately 594.4 square kilometres) falling within this distance threshold. This represents 9.22% of Chicago's total land area. Using the population density method, approximately 251,000 residents are estimated to live within a 10-minute straight-line walk of parks when physical barriers are not considered.

The increase from 400-meter to 800-meter buffers resulted in an additional 222.5 square kilometres of accessible area, representing a 60% increase in coverage. This demonstrates that extending the acceptable walking distance from 5 minutes to 10 minutes substantially expands potential park accessibility across the city

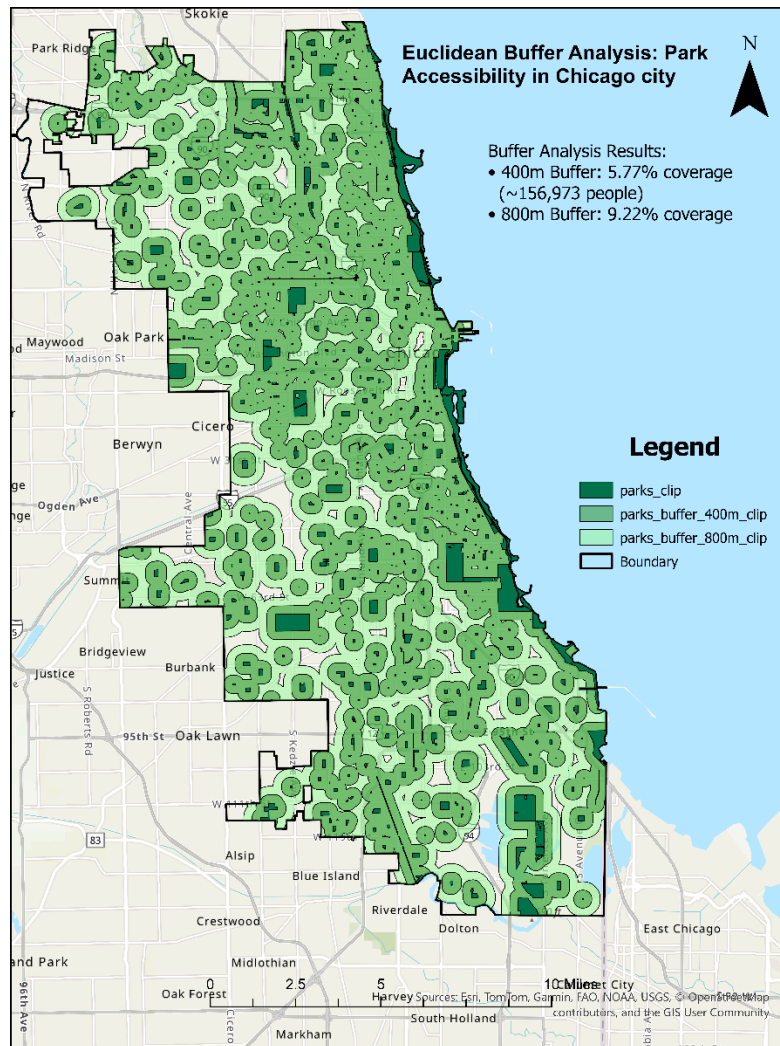


Figure 1: Buffer Analysis Map

Figure 1. Euclidean Buffer Analysis showing 400m and 800m buffer zones around parks in Chicago. The map illustrates proximity-based accessibility, with darker green showing parks, medium green showing 400m buffers (5-minute walk), and lighter green showing 800m buffers (10-minute walk). The analysis reveals that 5.77% and 9.22% of Chicago's land area falls within 400m and 800m of parks respectively.

Network Analysis Findings

The network-based service area analysis produced markedly different results compared to buffer analysis. Using an 800-meter walking distance along streets, the analysis identified 372,535,639

square meters (approximately 372.5 square kilometers) as accessible, representing 5.78% of Chicago's total land area. This figure is substantially lower than the 9.22% identified through 800-meter buffer analysis at the same distance threshold.

The network analysis revealed that street network configuration, physical barriers, and the absence of direct pedestrian connections reduce effective park accessibility by approximately 37% compared to straight-line distance assumptions. Applying population density calculations, an estimated 157,000 residents have realistic pedestrian access to parks within an 800-meter walk via streets, significantly fewer than the 251,000 suggested by buffer analysis

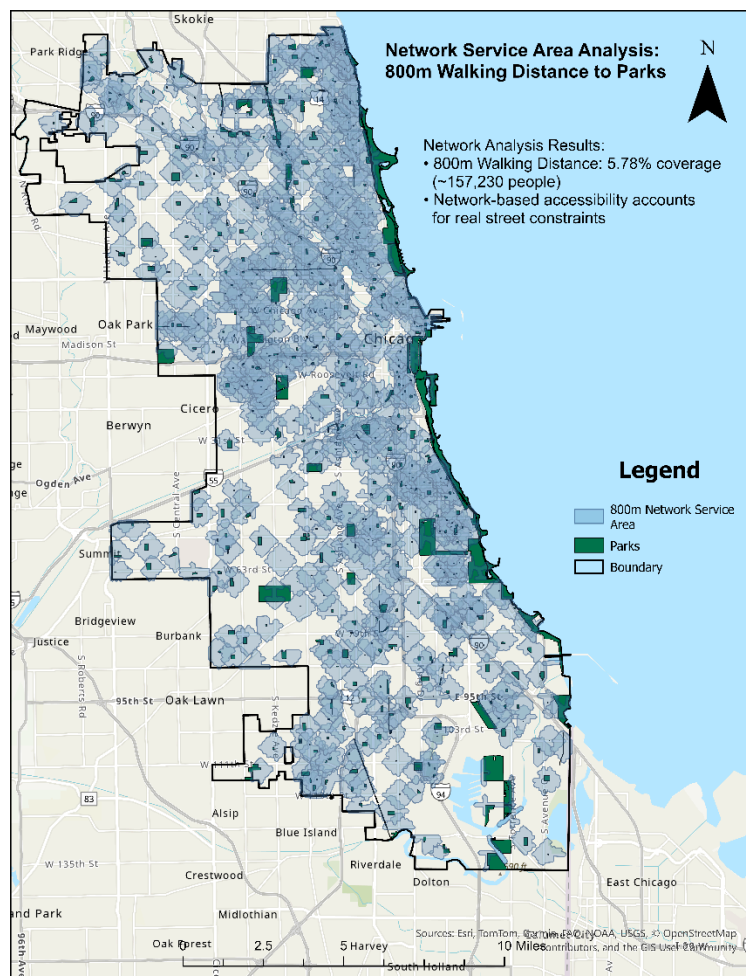


Figure 2: Network Service Area Map

Figure 2. Network Service Area Analysis displaying 800m walking distance accessibility to parks via the street network in Chicago. The map shows actual pedestrian-accessible areas (blue

shading) accounting for street connectivity and physical barriers. Dark green areas represent parks. White areas indicate locations where parks cannot be reached within 800m of walking along streets, demonstrating how street network configuration limits real-world accessibility compared to straight-line distance.

Comparative Analysis of Methods

Comparing the two methodologies reveals important insights about measuring urban accessibility. The 800-meter buffer analysis overestimated accessibility by approximately 60% compared to the network analysis (594.4 square kilometres versus 372.5 square kilometers). This substantial difference underscores the importance of considering actual pedestrian infrastructure when evaluating park access.

A particularly noteworthy finding emerged when comparing network-based 800-meter results to the 400-meter buffer results. The network analysis at 800 meters (372.5 square kilometers) produced nearly identical coverage to the 400-meter straight-line buffer (371.9 square kilometers). This suggests that in Chicago's urban environment, actual walking distances along streets are approximately double the straight-line distances, a finding consistent with urban form research indicating that street grid patterns, physical obstacles, and routing constraints significantly increase travel distances compared to direct-line measurements.

The population accessibility estimates further emphasize these disparities. While 400-meter and 800-meter network analyses both estimate approximately 157,000 residents with park access, the 800-meter buffer suggests 251,000 residents have access—a 60% overestimation. These differences have significant implications for urban planning decisions and resource allocation if accessibility is assessed using proximity measures alone without considering actual pedestrian routing

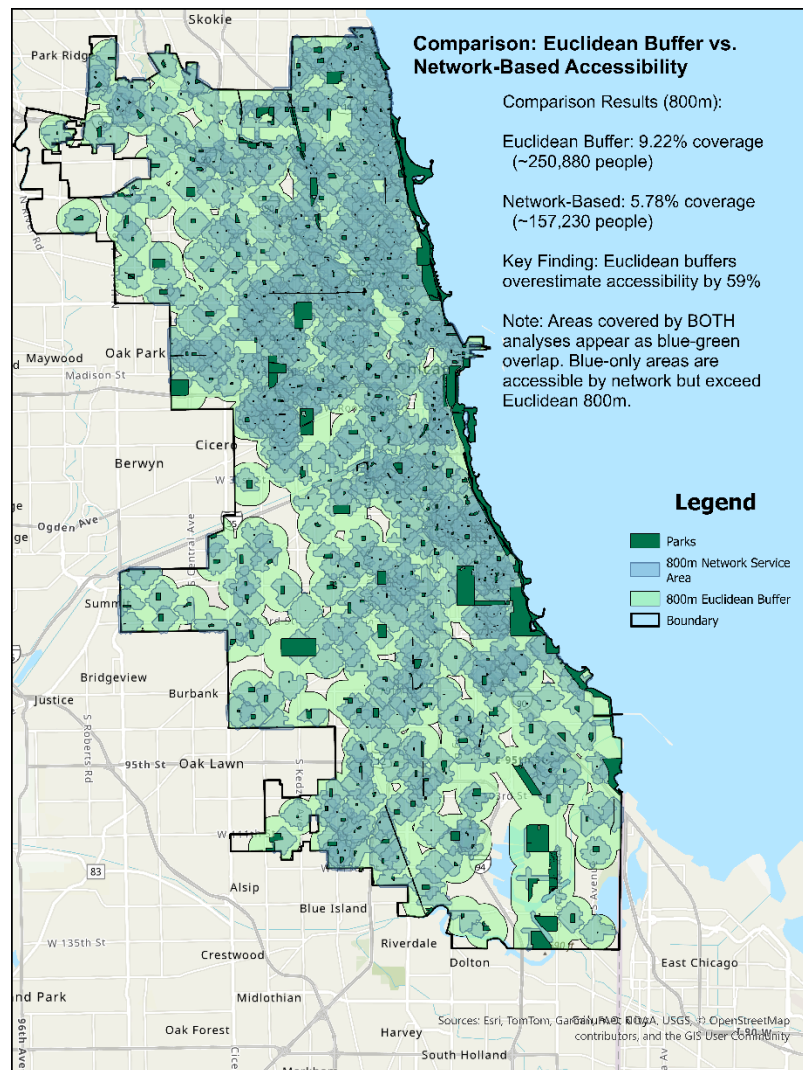


Figure 3: Comparison Map

Figure 3. Comparative analysis overlaying 800m Euclidean buffer zones (light green) and network-based service areas (blue) to illustrate differences between proximity and accessibility. Dark green areas represent parks. Blue-green overlap shows areas accessible by both methods. Blue-only areas demonstrate locations accessible via street networks but beyond 800m straight-line distance. Light green areas without blue indicate locations within 800m Euclidean distance but inaccessible within 800m of actual walking via streets. The map visually demonstrates that buffer analysis overestimates accessibility by 59% compared to network-based analysis.

Conclusion

This analysis demonstrates that less than 6% of Chicago's land area and approximately 157,000 residents (representing 5.8% of the city's population) have realistic pedestrian access to parks and green spaces within an 800-meter walk via the street network. While buffer analysis initially suggested that over 9% of the city falls within 800 meters of parks, network analysis revealed that street configurations, physical barriers, and pedestrian infrastructure constraints significantly limit actual accessibility.

The findings highlight a critical distinction between geographic proximity and functional accessibility. A park may be located relatively close to a residential area when measured in straight-line distance, yet remain functionally inaccessible if the street network does not provide direct pedestrian connections. Physical barriers such as highways, rivers, industrial areas, or gaps in the sidewalk network create substantial obstacles that buffer analysis cannot capture but that profoundly affect whether residents can actually reach green spaces.

This research has important implications for urban planning and public health initiatives. Expanding Park access requires a dual approach: creating new green spaces in underserved areas while simultaneously ensuring that adequate pedestrian infrastructure connects neighborhoods to existing parks. Investments in sidewalk networks, pedestrian crossings, and traffic calming measures may be as important as new park development in improving actual accessibility.

Several limitations should be acknowledged. First, the use of citywide population density rather than block-level demographic data limits the precision of population estimates. Future research incorporating detailed census data would enable identification of specific neighborhoods and demographic groups most affected by limited park access, particularly in relation to socioeconomic factors and environmental justice concerns. Second, this analysis measured accessibility solely in terms of distance and did not account for park size, quality, amenities, or maintenance conditions—factors that substantially influence whether residents actually use accessible green spaces. Third, the analysis assumed uniform pedestrian travel speeds and did

not account for variations related to age, mobility limitations, or other factors that affect walking ability.

Future research directions should address these limitations through several approaches. Incorporating detailed demographic data would enable equity analyses examining whether park accessibility varies systematically across income levels, racial and ethnic groups, or other demographic characteristics. Three-dimensional network analysis accounting for elevation changes and topographic barriers would provide more accurate accessibility estimates, particularly in areas with significant terrain variations. Quality assessments of accessible parks could evaluate whether residents have access not just to any green space but to well-maintained parks with desired amenities. Finally, temporal analyses examining how accessibility patterns have changed over time could inform understanding of whether urban development and planning interventions have improved or worsened park access equity.

Understanding these spatial patterns and accessibility disparities can inform evidence-based urban planning decisions that prioritize equitable park access for all residents. As cities face increasing population density, climate change impacts, and public health challenges, ensuring that all residents can reach quality green spaces within walking distance becomes ever more critical for promoting environmental justice, public health, and urban livability.

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